1.Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

Packet No.	Protocol	Info
15	DNS	Standard query A www.spotify.com
16	DNS	Standard query response A www.spotify.com → **35.186.224.25**
Packet No.	Protocol	Info
25	TLS (HTTPS)	Client Hello to www.spotify.com
26	TLS (HTTPS)	Server Hello from www.spotify.com

2.Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

```
C:\Users\helLO>nslookup www.flipkart.com
Server:  reliance.reliance
Address:  2405:201:2033:6025::c0a8:1d01

Non-authoritative answer:
Name:    flipkart.com
Address:  103.243.32.90
Aliases:  www.flipkart.com
```

3.Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

HTTP	HTTPS
Data is not encrypted.	Data is encrypted using SSL/TLS.
Less secure.	More secure and protects user information.
Uses Port 80.	Uses Port 443.

4.Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.

Step	Packet	Description
1	Standard query	The browser requests the IP address of **www.wikipedia.org**.
2	Standard query response	The DNS server replies with the IP address of **www.wikipedia.org**.

5.Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Protocol: DHCP (Dynamic Host Configuration Protocol)

Scenario:

When I visit a café and connect my phone to the free Wi-Fi, the **DHCP server** automatically assigns my phone an IP address, subnet mask, default gateway, and DNS server. After receiving these network settings, I can open **Instagram** and browse photos and videos without manually configuring any network settings.

Explanation:
DHCP makes joining a network simple by automatically providing all the necessary network configuration to the device.